	understood.	
6(a)	Isotopes are nuclides with the <u>same number of protons but different number of neutrons</u>.	[1] for correct answer
6(b)(i)	The binding energy of a nucleus <u>is the minimum amount of energy required to break the nucleus into its constituent particles.</u> <u>OR</u> The binding energy of a nucleus <u>is the energy released when a nucleus is formed from its constituent particles.</u>	[1] for correct answer
6(b)(ii)	Binding energy of tritium $=3(2.83)=8.49$ MeV $8.49\times10^{6}\times1.60\times10^{-19}$ $=(1.00783u+2\times1.00867u-m)(3.00\times10^{8})^{2}$ $m=3.01608\ u$ *Common mistakes:*	[1] for BE of tritium [1] for correct substitution [1] for correct answer

No.	Solution	Remarks
	- *mass of proton + mass of 2 neutrons = mass of tritium nucleus* - *implying through calculations that mass of tritium nucleus > mass pf proton + mass of 2 neutrons* - *not multiplying the given 2.83 MeV by 3 to obtain the binding energy of a tritium nucleus.*	
6(c)(i)	${}^{2}_{1}\text{H} + {}^{2}_{1}\text{H} \rightarrow {}^{3}_{1}\text{H} + {}^{1}_{1}\text{H}$	[1] for correct answer
6(c)(ii)1.	BE of tritium – 2 × BE of deuterium = 4.03 MeV 8.49 – 2 × BE of deuterium = 4.03 MeV BE of deuterium = 2.23 MeV *Common mistake:* - *binding energy of the products is smaller than the binding energy of the reactants.*	[1] for correct substitution [1] for correct answer
6(c)(ii)2.	No. of reactions required per unit time $= \dfrac{2.86 \times 10^{9}}{4.03 \times 10^{6}(1.6 \times 10^{-19})}$ Mass of deuterium required per unit time $= \left[\dfrac{2.86 \times 10^{9}}{4.03 \times 10^{6}(1.60 \times 10^{-19})}\right](2 \times 2 \times 1.66 \times 10^{-27})$ $= 2.95 \times 10^{-5}$ kg	[1] for no. of reactions [1] for correct substitution [1] for correct answer
7(a)	It is impossible to determine exactly which nucleus and exactly	[1] for which